

Indian Institute of Technology Kanpur

Telecom Customer Churn Prediction

Using Data Mining Techniques

Project Members:

Vishnu Vardhan Vadthya (231080109)
Anshika Rai (230162)
Sejal Dubey (241080092)
Shekhar Suman (241080093)
Shivam Kumar Gond (241080094)

Instructor: Prof. Amit Mitra sir

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Chapter 1

Introduction

1.1 Background and Motivation

Customer churn prediction is a crucial problem in industries such as telecommunications, banking, and subscription services, where retaining existing customers is more cost-effective than acquiring new ones. Churn leads to significant revenue loss and reflects underlying issues in customer satisfaction and service quality. Real-world customer data is often high-dimensional, imbalanced, and exhibits non-linear relationships between variables such as usage behavior and customer demographics. This project is motivated by the need for flexible, data-driven approaches that can effectively model such complexities. Machine learning and data mining techniques provide robust tools to identify at-risk customers and improve predictive performance, making them well-suited for churn analysis.

1.2 Data Mining Perspective

Customer churn prediction can be viewed as a data mining problem, where the goal is to extract useful patterns from large and complex customer datasets. These datasets typically contain demographic information, usage behavior, and transaction records, forming a high-dimensional feature space.

From a methodological perspective, churn prediction is a supervised classification task that aims to learn the relationship between input features X and the churn outcome Y. However, real-world data often exhibits challenges such as class imbalance, missing values, and non-linear relationships.

Data mining techniques provide flexible and robust approaches to handle these complexities, enabling accurate prediction and identification of key factors influencing customer churn.

1.3 Problem Statement

This study applies data mining techniques to the customer churn problem, aiming to identify patterns in customer behavior that lead to service discontinuation. Initial analysis focuses on assessing data quality issues such as class imbalance, missing values, and non-linear relationships to justify the choice of flexible modeling approaches.

The analysis then addresses key predictive questions, including the impact of customer usage patterns, service features, and demographic factors on churn (classification framework). Finally, multiple machine learning models are compared to evaluate their predictive performance and to identify the most influential features driving customer attrition, highlighting the effectiveness of data mining techniques in capturing complex behavioral patterns.

Chapter 2

Objectives

This project aims to analyze and predict customer churn using data mining techniques. The objectives include:

1. Evaluating data characteristics such as missing values, and non-linearity to justify the use of machine learning methods.
2. Applying classification algorithms (Logistic Regression, Decision Tree, Random Forest, KNN, SVM, MLP) to predict customer churn based on usage, billing, and demographic features.
3. Assessing the impact of key features (e.g., *tenure, monthly charges, contract type*) on churn behavior through feature importance analysis.
4. Comparing model performance using evaluation metrics such as Accuracy, Precision, Recall, and F1-score.
5. Identifying the most effective model and extracting insights to support customer retention strategies.

2.1 Dataset Description

2.1.1 Origin and Provenance

The Customer Churn dataset contains information about customer behavior in a telecommunications company. It consists of 7043 customer records with 21 variables, capturing demographic details, service usage, and account-related information. The dataset is widely used as a benchmark for classification and data mining tasks in churn prediction.

2.1.2 Data Dictionary

The dataset includes multiple predictor variables and one target variable (Churn). The variables can be grouped as follows:

Variable	Description	Type
customerID	Unique customer identifier	Categorical
gender	Customer gender	Categorical
SeniorCitizen	Indicates if customer is a senior citizen (1 = Yes, 0 = No)	Binary
Partner	Whether the customer has a partner	Binary
Dependents	Whether the customer has dependents	Binary
tenure	Number of months the customer has stayed with the company	Continuous
PhoneService	Whether the customer has phone service	Binary
MultipleLines	Whether the customer has multiple lines	Categorical
InternetService	Type of internet service used	Categorical
OnlineSecurity	Whether online security service is provided	Binary
OnlineBackup	Whether online backup service is provided	Binary
DeviceProtection	Whether device protection is provided	Binary
TechSupport	Whether tech support is provided	Binary
StreamingTV	Whether streaming TV service is used	Binary
StreamingMovies	Whether streaming movie service is used	Binary
Contract	Type of customer contract	Categorical
PaperlessBilling	Whether paperless billing is enabled	Binary
PaymentMethod	Mode of payment used by customer	Categorical
MonthlyCharges	Monthly charges incurred by the customer	Continuous
TotalCharges	Total charges incurred by the customer	Continuous
Churn	Whether the customer churned (Target variable)	Binary

Table 2.1: Data Dictionary for Customer Churn Dataset

Chapter 3

Exploratory Data Analysis

3.1 Univariate Analysis

The gender distribution is nearly balanced, with both categories contributing approximately 50% of the dataset, indicating no significant bias.

The distribution *Churn Rate* is imbalanced, with a larger share of non-churn customers compared to churn customers.

Service-related features such as *PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, and StreamingMovies* show varying levels of engagement. Basic services are widely adopted, while value-added services have lower subscription rates.

The distribution of *tenure* exhibits a bimodal pattern, suggesting the presence of both short-term and long-term customers.

The overall distribution of *Churn* indicates that most customers do not churn, while a smaller proportion exits the service, reflecting class imbalance.

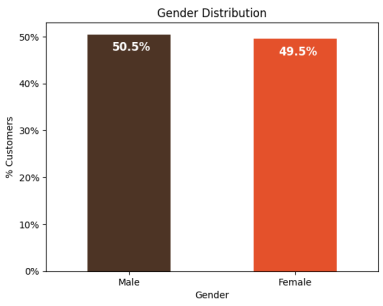


figure: Gender Distribution

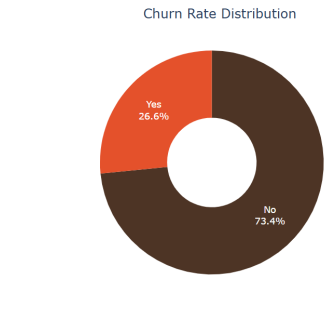


figure: Churn rate Distribution

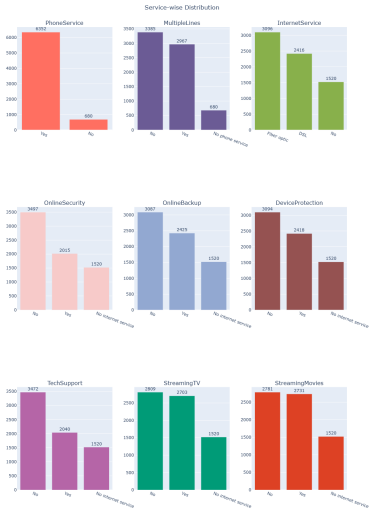


figure: Service Distributions

3.2 Bivariate Analysis

The relationship between *tenure* and *churn* shows a strong inverse association, with customers having lower tenure more likely to churn. This indicates that churn is more prevalent among newer customers.

Payment method analysis reveals that customers using electronic checks exhibit a higher tendency to churn compared to those using bank transfers or credit cards.

The KDE plots of *MonthlyCharges* and *TotalCharges* suggest that customers with higher monthly charges and lower total charges are more likely to churn, indicating price sensitivity and shorter service duration.

The relationship between *TechSupport* and churn shows that customers without technical support have higher churn rates, while those with support are more likely to stay, emphasizing its role in retention.

Churn distribution across gender remains nearly identical, confirming that gender is not a significant factor influencing churn.

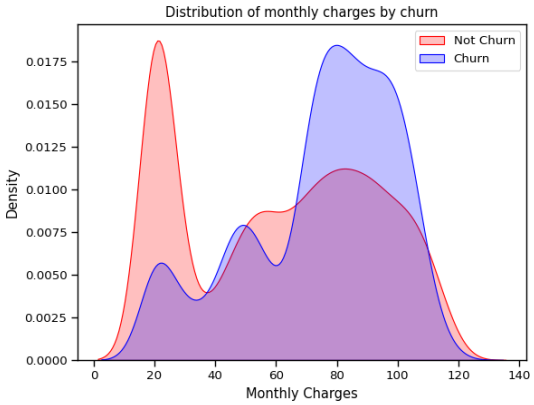


figure: Monthly Charges vs Churn

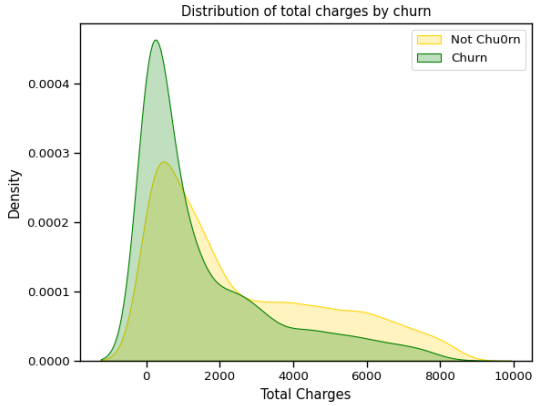
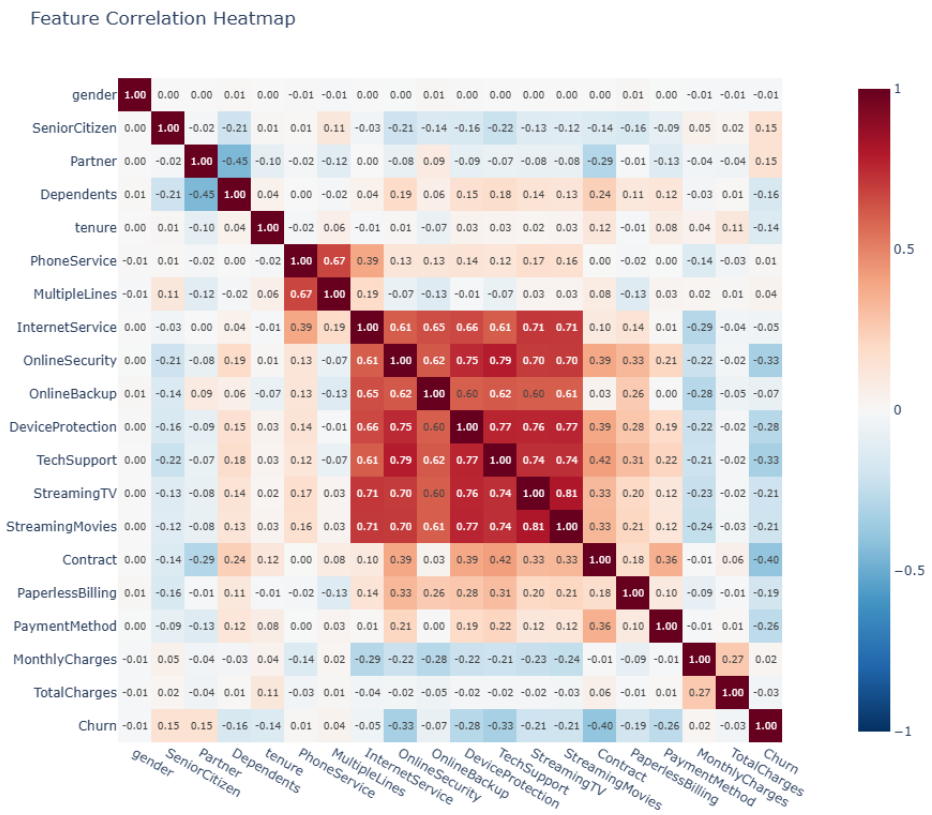


figure: Total Charges vs Churn

3.3 Correlation Analysis

The correlation heatmap highlights key relationships between features and churn. *tenure* and *Contract (Two year)* show strong negative correlation with churn, indicating better retention among long-term customers. In contrast, *InternetService (Fiber optic)* and *MonthlyCharges* are positively correlated with churn, suggesting higher churn among these customers.

Demographic variables like *gender* show negligible impact, while *SeniorCitizen* has a weak positive correlation. Overall, tenure, contract type, and pricing-related features emerge as major factors influencing churn.



Chapter 4

Methodology

4.1 Data Preprocessing

Data preprocessing is a crucial step in the data mining pipeline to ensure data quality and improve model performance. The customer churn dataset contains a mix of categorical and numerical variables, requiring appropriate transformation before model implementation.

4.1.1 Data Cleaning

The dataset was examined for missing values and inconsistencies. The variable `TotalCharges`, originally stored as a string, was converted to a numerical format. Any missing or invalid entries were handled appropriately to ensure consistency.

4.1.2 Handling Missing Data

The *tenure* variable had 11 missing values, which were removed due to their negligible proportion. The distribution remained unchanged after removal.

4.1.3 Feature Scaling

Numerical features such as `tenure`, `MonthlyCharges`, and `TotalCharges` were scaled using standardization to ensure that all features contribute equally to the model:

$$Z = \frac{X - \mu}{\sigma}$$

4.1.4 Train-Test Split

The dataset was split into training and testing sets in the ratio of 70:30 to evaluate model performance on unseen data. This ensures that the models generalize well and do not

overfit the training data.

4.2 Feature Engineering

Feature engineering is an essential step in improving model performance by transforming raw data into meaningful input features. In this study, relevant transformations and selections were applied to enhance the predictive power of the models.

4.2.1 Feature Selection

Irrelevant and redundant features were identified and removed to reduce noise and improve model efficiency. In particular, the `customerID` variable was excluded as it does not contribute to prediction.

4.2.2 Encoding of Categorical Variables

Categorical features like `'gender'`, `'Partner'`, `'Dependents'`, `'PhoneService'`, `'MultipleLines'`, `'InternetService'`, `'OnlineSecurity'`, `'OnlineBackup'`, `'DeviceProtection'`, `'TechSupport'`, `'StreamingTV'`, `'StreamingMovies'`, `'Contract'`, `'PaperlessBilling'`, `'PaymentMethod'`, `'Churn'` were transformed into numerical representations using encoding techniques such as one-hot encoding and label encoding. This ensures compatibility with machine learning algorithms while preserving meaningful information.

4.2.3 Transformation of Numerical Features

Numerical variables such as `tenure`, `MonthlyCharges`, and `TotalCharges` were examined for skewness and distribution. Appropriate transformations and scaling were applied to improve model stability.

4.2.4 Feature Importance Analysis

Feature importance techniques were used to identify key variables influencing churn. Variables such as `tenure`, `MonthlyCharges`, and `Contract` were found to have a significant impact on customer churn behavior.

4.2.5 Dimensionality Consideration

Care was taken to avoid the curse of dimensionality by selecting only relevant features and avoiding unnecessary expansion of feature space.

Chapter 5

Mathematical Formulation of Models

This chapter presents the mathematical foundations of the machine learning models used for customer churn prediction.

5.1 Logistic Regression

Logistic Regression models the probability of a binary outcome using the logistic function:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta^T X)}}$$

where X represents input features and β are model parameters.

5.2 Decision Tree

Decision Trees split data based on feature values using impurity measures such as Gini Index:

$$Gini = 1 - \sum_{i=1}^{k} p_i^2$$

where p_i is the probability of class i.

5.3 Random Forest

Random Forest is an ensemble of decision trees. The final prediction is obtained by averaging:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} T_b(X)$$

where T_b represents individual trees.

5.4 K-Nearest Neighbors (KNN)

KNN is a distance-based method. The Euclidean distance is commonly used:

$$d(x, x_i) = \sqrt{\sum_{j=1}^{p} (x_j - x_{ij})^2}$$

The class is assigned based on the majority of nearest neighbors.

5.5 Support Vector Machine (SVM)

SVM aims to find a hyperplane that maximizes the margin:

$$w^T x + b = 0$$

The optimization problem is:

$$\min \frac{1}{2}||w||^2 \quad \text{subject to } y_i(w^T x_i + b) \geq 1$$

5.6 Linear Discriminant Analysis (LDA)

LDA assumes Gaussian distributions and maximizes class separability:

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k$$

where μ_k is class mean and Σ is covariance.

5.7 Multi-Layer Perceptron (MLP)

MLP is a neural network model:

$$z = Wx + b$$

$$y = \sigma(z)$$

where σ is an activation function (e.g., sigmoid or ReLU).

5.8 Principal Component Analysis (PCA)

PCA reduces dimensionality by projecting data onto principal components:

$$Z = XW$$

where W contains eigenvectors of the covariance matrix.
The variance explained is maximized along these components.

Chapter 6

Model Building

6.1 Problem Formulation

The customer churn problem is formulated as a supervised classification task, where the input features X represent customer attributes and the target variable Y indicates whether a customer churns (1) or not (0).

6.2 Train-Test Split

The dataset is divided into training and testing sets to evaluate model performance on unseen data. Typically, a split of 70% training and 30% testing is used.

6.3 Models Implemented

The following machine learning models are implemented:

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Linear Discriminant Analysis (LDA)
- Multi-Layer Perceptron (MLP)

Each model is trained on the processed dataset and evaluated based on standard classification metrics.

6.4 Model Evaluation and Predictions

6.4.1 Metrics

$$Accuracy = \frac{TP + TN}{Total}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2PR}{P + R}$$

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.81	0.75	0.73	0.74
Decision Tree	0.70	0.63	0.63	0.63
Random Forest	0.78	0.72	0.68	0.69
KNN	0.76	0.69	0.70	0.70
SVM	0.80	0.74	0.69	0.71
LDA	0.80	0.74	0.72	0.73

Table 6.1: Comparison of Model Performance

6.5 Model-wise Analysis

6.5.1 Logistic Regression

Logistic Regression achieves the highest accuracy (81%) among all models. It performs well in identifying non-churn customers, as seen from the high recall (0.89) for class 0. However, the recall for churn customers (class 1) is relatively lower (0.57), indicating that some churn cases are misclassified as non-churn.

6.5.2 Decision Tree

Decision Tree shows the lowest performance (70% accuracy). Although it captures non-linear relationships, it suffers from overfitting and poor generalization, particularly in identifying churn customers (recall = 0.48).

6.5.3 Random Forest

Random Forest improves over Decision Tree with better generalization (78% accuracy). However, it still struggles to identify churn customers effectively, with a relatively low

recall (0.46) for class 1.

6.5.4 K-Nearest Neighbors (KNN)

KNN provides moderate performance (76% accuracy). It maintains a balanced recall for both classes but is affected by high dimensionality and feature scaling issues.

6.5.5 Support Vector Machine (SVM)

SVM achieves strong performance (80% accuracy) by maximizing class separation. However, it shows lower recall (0.47) for churn customers, indicating difficulty in identifying minority class instances.

6.5.6 Linear Discriminant Analysis (LDA)

LDA performs comparably to Logistic Regression (80% accuracy) and shows a balanced performance across classes. It provides relatively better recall (0.57) for churn customers compared to several other models.

6.5.7 Multi-Layer Perceptron (MLP)

The MLP classifier achieves an accuracy of 79%, which is comparable to SVM and LDA. It performs well in identifying non-churn customers, with a high recall of 0.93 for class 0. However, its performance on churn customers is relatively weak, with a recall of only 0.40.

This indicates that the model fails to correctly identify a significant portion of churn cases (false negatives), which is a critical limitation in churn prediction tasks.

Although MLP is capable of capturing complex non-linear relationships, its performance may be affected by factors such as hyperparameter tuning, network architecture, and class imbalance in the dataset.

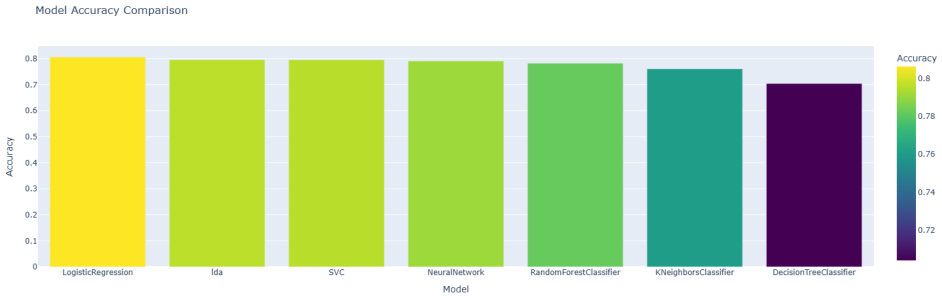


Figure 6.1: Model Comparison Plot

Key Observations

- Logistic Regression and LDA provide the best overall performance in terms of accuracy and balance.
- Decision Tree performs the worst due to overfitting and poor generalization.
- Random Forest improves stability but still struggles with minority class detection.
- MLP accuracy appears high but is affected by class imbalance.
- Performance can be improved with better tuning and balancing techniques.
- Most models show lower recall for churn customers, indicating difficulty in identifying the minority class

6.6 Dimensionality Reduction using PCA

To reduce dimensionality and remove redundancy among features, Principal Component Analysis (PCA) was applied. The dataset was transformed into 15 principal components while retaining 95% of the variance.

- PCA was fitted on the training data.
- The same transformation was applied to the test data.

6.6.1 Model Performance after PCA

Model	Accuracy (After PCA)
Logistic Regression	0.799
LDA	0.798
SVM	0.795
MLP	0.791
Random Forest	0.781
KNN	0.769
Decision Tree	0.731

Table 6.2: Model Performance after Applying PCA

Key Observations:

- After applying PCA, the feature set was reduced from 31 to 15 while retaining 95% of the variance, with almost no loss in model accuracy.
- Logistic Regression and LDA show nearly identical performance before and after PCA.

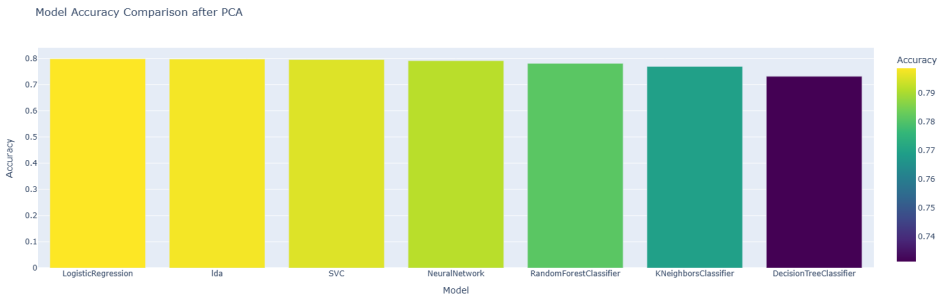


Figure 6.2: Model Comparison Plot after PCA

- Tree-based models such as Random Forest do not benefit from PCA, as they handle feature interactions internally.
- Slight reduction in recall for churn cases is observed in some models, indicating loss of important feature information.
- PCA reduces dimensionality and computational complexity but may decrease interpretability.

6.7 Comparison: Before vs After PCA

Observation:

- Models such as Logistic Regression and LDA show negligible change.
- Tree-based models (Random Forest) do not benefit from PCA.
- PCA slightly improves stability but may reduce interpretability.

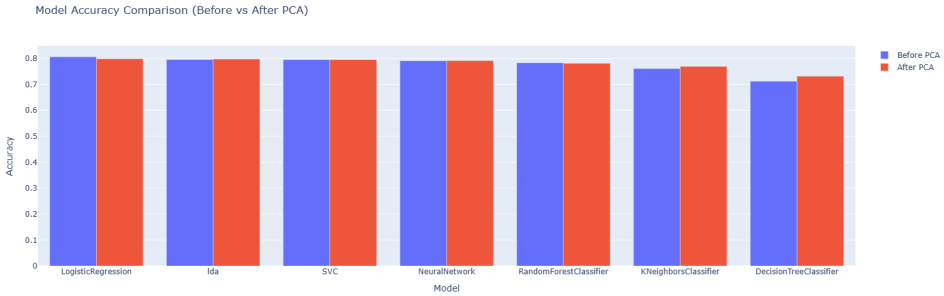


Figure 6.3: Model Comparison: Before vs After PCA

6.8 Business Interpretation

The results show that while models achieve good accuracy, identifying churn customers remains challenging. From a business perspective, false negatives are critical, as missed churn cases lead to revenue loss.

Key factors such as tenure, monthly charges, and contract type influence churn behavior. Focusing on improving recall can help organizations identify at-risk customers early and implement effective retention strategies.

6.9 Conclusion

This project applied data mining techniques to predict customer churn using various machine learning models. The analysis revealed that the dataset contains class imbalance and complex relationships, making churn prediction a challenging task.

Among the models, Logistic Regression and LDA showed strong overall performance, while other models captured non-linear patterns effectively. However, a common limitation across models is the relatively low recall for churn customers, indicating difficulty in identifying all at-risk customers.

Overall, the study highlights that improving recall for churn cases is more important than achieving high accuracy, as missed churn predictions have significant business implications.

6.10 Future Scope

This study establishes a baseline framework for predicting customer churn, which can be further enhanced through advanced techniques.. However, several enhancements can be explored to further improve model performance, interpretability, and real-world applicability.

- **Hyperparameter Tuning:** Hyperparameter tuning (using Grid Search or Random Search) improves model performance by optimizing settings like tree depth, number of estimators, and regularization, leading to better generalization and higher recall in churn prediction..
- **Advanced Machine Learning Models:** Ensemble methods like Gradient Boosting (e.g., XGBoost, LightGBM) can improve performance by capturing complex non-linear patterns and often outperforming traditional classification models.
- **Deep Learning Approaches:** Artificial Neural Networks (ANNs) and advanced deep learning models can capture complex feature interactions and improve performance, especially with well-tuned models and high-dimensional data.

- **Handling Class Imbalance:** Techniques like SMOTE, undersampling, and cost-sensitive learning help handle class imbalance and improve recall for churn prediction.
- **Feature Engineering Enhancements:** Feature engineering (e.g., customer lifetime value, average spending, interaction terms) helps capture hidden patterns and improve model performance.
- **Threshold Optimization:** Adjusting the classification threshold (instead of the default 0.5) can improve recall for churn customers, which is crucial for business needs.
- **Real-Time Deployment:** Deploying the model in a real-time system enables continuous monitoring, early detection of at-risk customers, and proactive retention actions.

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